

Munim Thahmid

FORMAL METHODS / SOFTWARE SYSTEMS / RESEARCH ENGINEERING

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RESEARCH FOCUS

Formal methods and dependable software systems, with current work on AI-assisted TLA+ proof evaluation, scalable TLAPS verification, and reliable agentic-SRE benchmarks.

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Computer Science and Engineering

Jan 2022 - May 2026

CGPA: 3.46 / 4.00 | Dhaka, Bangladesh

RESEARCH EXPERIENCE

Research Intern | University of Illinois Urbana-Champaign

May 2026 - Present

Remote | Advisor: Prof. Tianyin Xu

- Active contributor to TLAPS-Bench, an evaluation framework for AI-generated TLA+ proofs; authored seven merged pull requests across proof checking and evaluator infrastructure.
- Scaled verification of large proof artifacts through TLAPM fingerprint-cache reuse, fail-fast validation, and parallel sharding at top-level theorem boundaries.
- Improved benchmark validity and resilience with continuation runs, infrastructure-failure retries, result deduplication, and explicit model reasoning controls.
- Built a reproducible SREGym scenario for Kubernetes node conntrack exhaustion, including runtime setup, oracle logic, and analysis of blocked diagnostic-access paths.

Undergraduate Researcher | BUET

2025 - 2026

Supervisor: Dr. Sadia Sharmin

- Led layer-wise probing of Bengali phone-like units across Whisper-small, Whisper-medium, and Whisper-large-v3 under speaker-disjoint evaluation.
- Validated representation trends with XLS-R comparison, ABX discriminability, confidence filtering, duration stratification, and probe ablations; work accepted to INTERSPEECH 2026.

ENGINEERING EXPERIENCE

Software Engineer | Yobo AI

Oct 2024 - Feb 2026

Remote | Intern, then part-time

- Built backend systems, testing infrastructure, and production features for AI voice-agent applications over more than one year.
- Developed automated Python testing infrastructure that reduced a regression cycle from roughly eight hours to fifteen minutes.

SELECTED PUBLICATION

Layer-wise Probing of Whisper's Encoder Representations for Bengali Phone-like Units

Munim Thahmid and Sadia Sharmin. Accepted to INTERSPEECH 2026.

TECHNICAL SKILLS

Formal methods: TLA+, TLAPS/TLAPM, machine-checkable proof evaluation

Programming & systems: Python, C/C++, TypeScript/JavaScript, Bash, Linux, Docker, Kubernetes, Git

ML & research: PyTorch, Hugging Face, Whisper, XLS-R, scikit-learn, pandas, NumPy, statistical evaluation